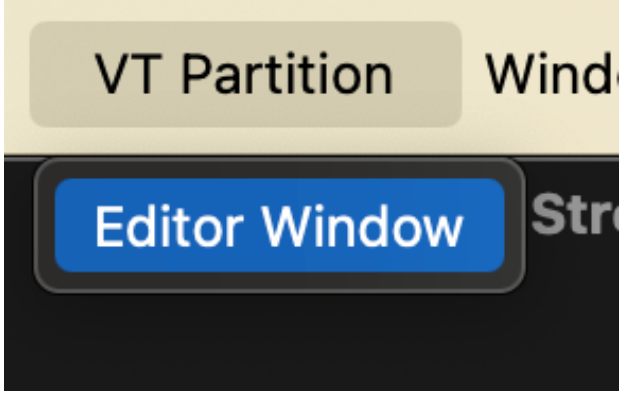
VTPartition Document

VTPartition use octree to voxelize the active scene. The result data is save as `.vtscene` file.

Octree is constructed by `VTCode`, which is a new method for representing voxels, similar to `Morton Code`. We use `32` bit `VTCode` to construct an octree, which can represent a `1024` unit range map. To represent a very large map, we use a group of octrees to divide the scene into grid.

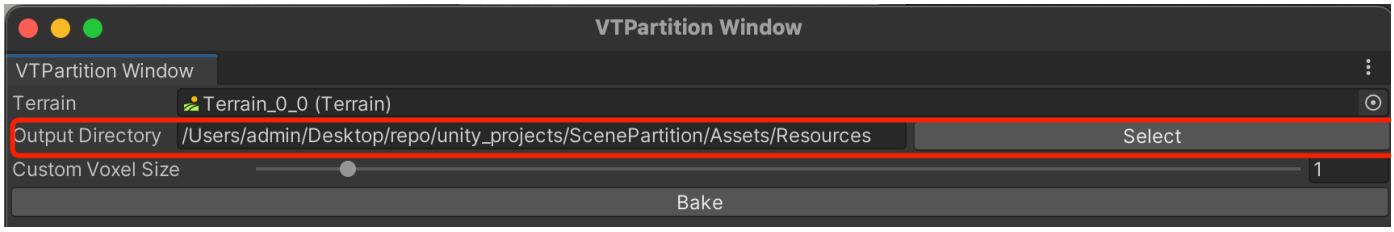
Bake Window

Click the top menu: `VTPartition - Editor Window`, to open the bake window.



The window show as below. And then, click `Select` button to select the output directory.

For example, we choose the `Resources` folder in project.



Set the `Custom Voxel Size` as a suitable value, Value `1` is recommended. This parameter mean the minimum voxel size in octree.

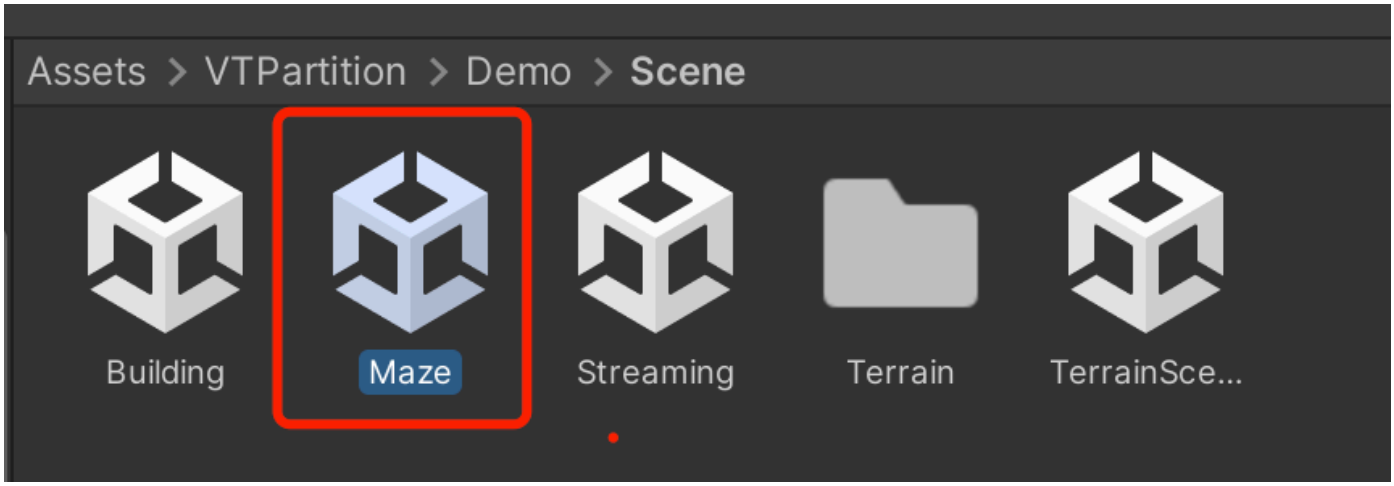
Click `Bake` button to bake the whole scene. The group size and scene size will be calculated automatically. The progress is show as below.

Demo Usage

Our demo show three scenes: `Maze`, `Building`, `TerrainScene`, `Streaming`

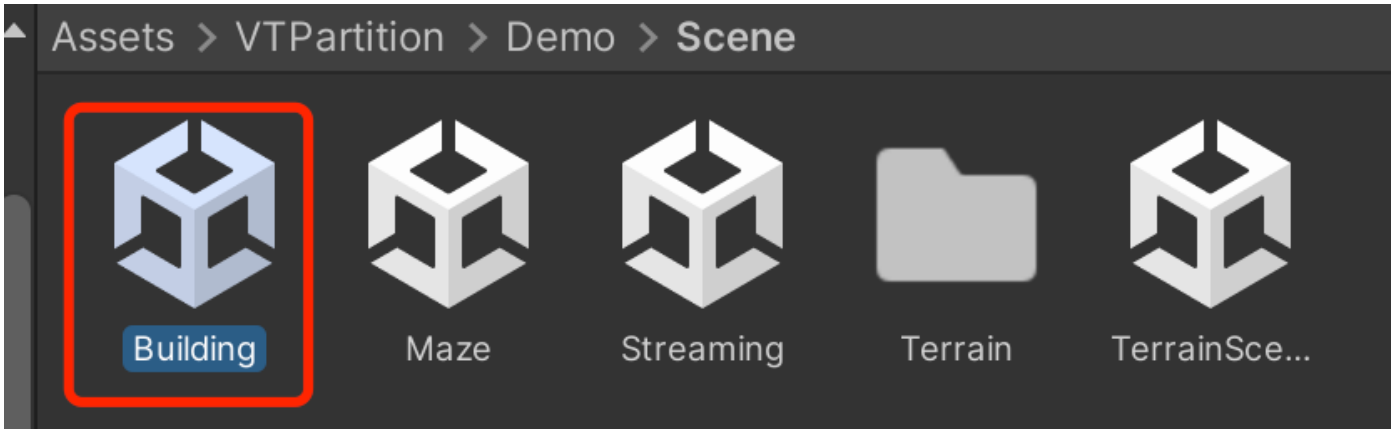
Maze Scene

Open `Maze` scene and play it. Use `w`, `s`, `a`, `d` to control camera to tranverse the scene.



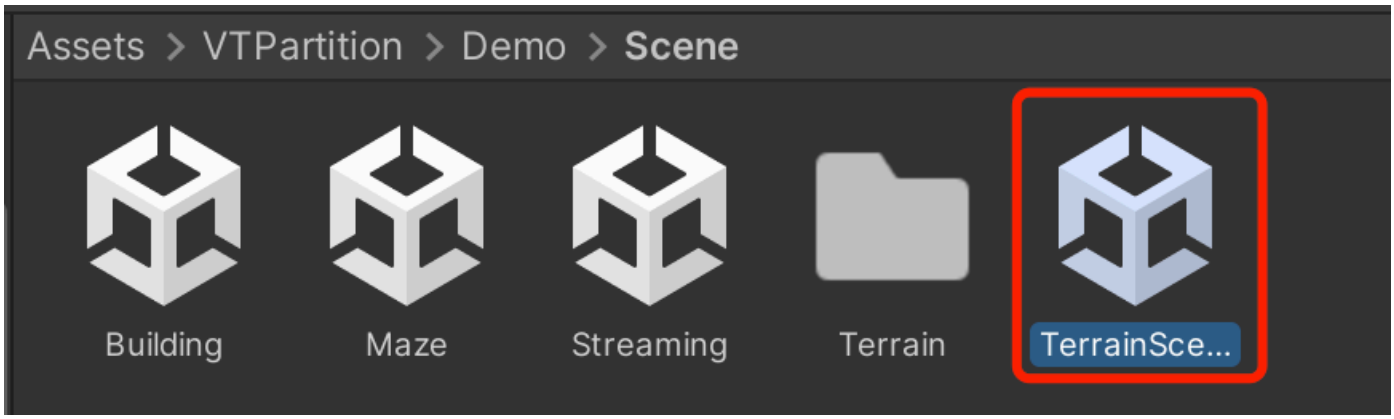
Building Scene

Open `Building` scene and play it. Use `w`, `s`, `a`, `d` to control camera to tranverse the scene.



TerrainScene

Open `TerrainScene` and play it. Use `w`, `s`, `a`, `d` to control camera to tranverse the scene.



Streaming Scene

Open `Streaming` scene. The `Player` agent will move automatically and the streaming loader will load the voxel scene within `load distance` and unload scene out of the `unload distance`.